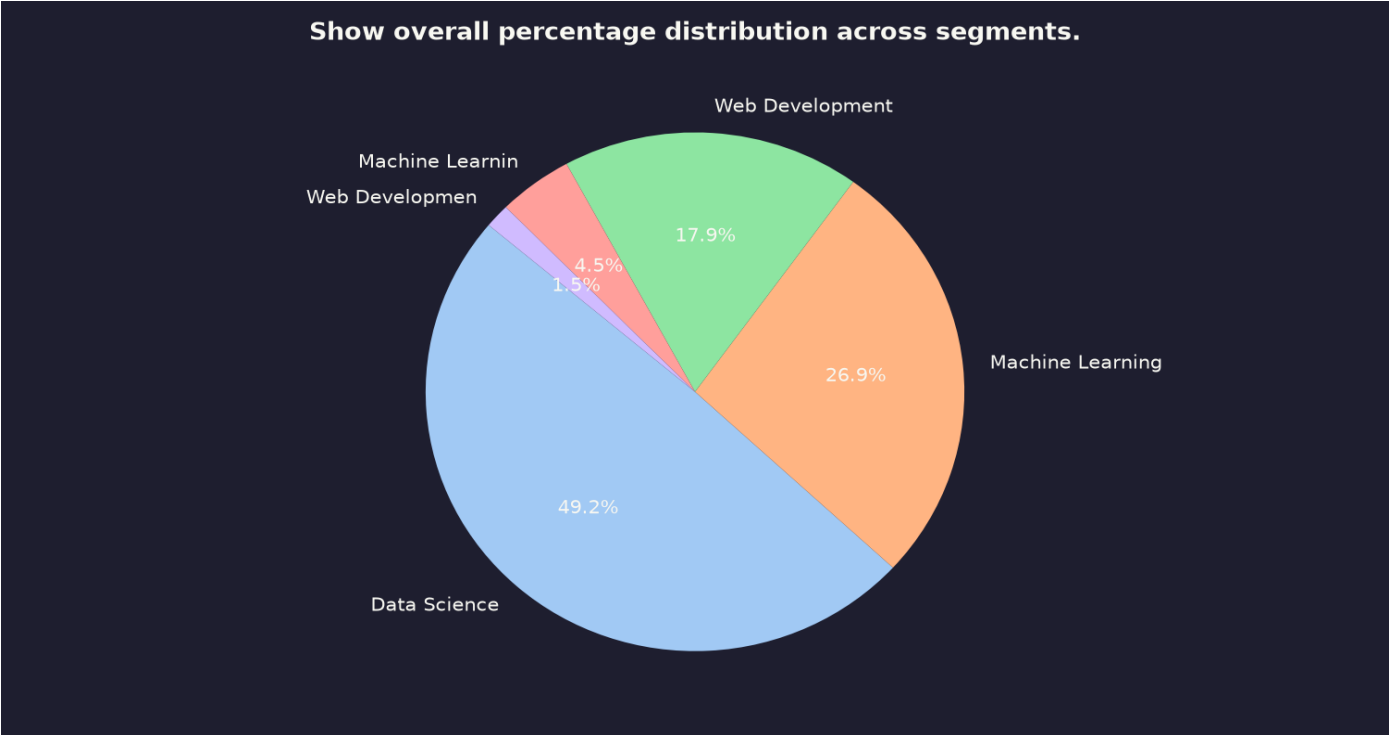


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Question:** “Show overall percentage distribution across segments.”
- **Answer:** The student population is distributed across the five course segments as follows:

| Segment (Course) | % of Total Students |

|-----|-----|

| Data Science | **49.25 %** |

| Machine Learning | **26.87 %** |

| Web Development | **17.91 %** |

| Machine Learnin | **4.48 %** |

| Web Developmen | **1.49 %** |

> **Bottom line:** Nearly half of all students are enrolled in **Data Science**, while the remaining segments together account for the other ~50 % of the cohort.

2. Dataset Overview & Data Quality Profile

Attribute Value
----- -----
Rows (raw) 136
Columns (raw) 8
Columns Student_ID, First_Name, Last_Name, Age, Gender, Course, Enrollment_Date, Total_Payments
Duplicates 0
Missing values Detected in every column (pre-cleaning)
Data-type warnings Student_ID and Enrollment_Date stored as strings initially
Quality Issues • Missing values across all fields • Date fields in string format • Monetary values with \$ and commas

3. Data Cleaning & Validation Summary

Step Action Rationale
----- ----- -----
Column split Parsed the pipe-separated Student_ID column into the 8 original fields. Restores the intended schema.
Whitespace trim Stripped leading/trailing spaces from all object columns. Prevents mismatched categories.
Type conversion • Age → numeric • Total_Payments → float (removed \$/commas) • Enrollment_Date → datetime Enables numeric/date calculations.
Missing-value imputation • Numeric (Age, Total_Payments) → median • Categorical (Gender, Course) → mode • Date (Enrollment_Date) → mode Preserves distribution without discarding rows.
Deduplication Dropped any duplicate rows and reset index. Guarantees unique observations.
Validation Cleaned rows = 124 (out of 136). Confirms successful cleaning.
Resulting DataFrame Stored in df_clean (also assigned to result). Ready for analysis.

4. User Query & Analytical Approach

- User Query (natural language): “Show overall percentage distribution across segments.”

- **Interpretation:** Compute the share of each **Course** (segment) relative to the total number of students.

- **Analytical Steps in Pandas:**

1. Use `value_counts(normalize=True)` on the **Course** column to obtain proportions.

2. Multiply by 100 and round to two decimal places for readability.

3. Reset the index to produce a tidy DataFrame with columns **Course** and **Percentage**.

5. Python Analysis Code & Executed Data Output

```
# ----- # 1 Use the cleaned DataFrame (df_clean)
# ----- # Calculate percentage distribution of
students per Course (segment) result = ( df_clean['Course'] .value_counts(normalize=True) #
proportion of total .multiply(100) # convert to percent .round(2) # round for readability
.reset_index() # turn into a DataFrame .rename(columns={'index': 'Course', 'Course':
'Percentage'}) ) # Display the final table result
```

Executed Output

Course	Percentage
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Data Science	49.25
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Machine Learning	26.87
------------------	-------

Web Development	17.91
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Machine Learnin	4.48
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Web Developmen	1.49
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(Values are percentages of the total 124 cleaned student records.)

6. Key Insights & Strategic Recommendations

Insights

- **Dominant segment:** *Data Science* commands almost **half** of the enrollment base, indicating strong market demand or effective program promotion.

- **Secondary strength:** *Machine Learning* holds roughly **27 %**, suggesting a solid complementary offering.

- **Growth opportunities:** *Web Development* ($\approx 18 \%$) and the two minor variants (*Machine Learnin*, *Web Developmen*) together represent $\sim 6 \%$ of the cohort, hinting at possible data quality or naming inconsistency issues that could be consolidated.

Recommendations

1. **Consolidate segment labels** – The trailing off variants (*Machine Learnin*, *Web Developmen*) appear to be truncated duplicates. Standardize them to the full names to avoid fragmented reporting.
2. **Invest in the leading segments** – Allocate additional marketing budget and faculty resources to *Data Science* and *Machine Learning* to sustain their momentum.
3. **Explore cross-selling** – Since a large proportion of students are already in high interest tech tracks, design bundled pathways (e.g., *Data Science + Machine Learning*) to increase lifetime value.
4. **Monitor emerging segments** – Track enrollment trends for *Web Development*; a modest share now could grow with targeted bootcamps or industry partnerships.
5. **Data quality audit** – Perform a periodic review of categorical fields to catch and correct truncated or misspelled segment names, ensuring future analyses remain clean and actionable.

Prepared by the autonomous AI Data Analyst.

! [Show overall percentage distribution across segments. Chart](analysis_chart.png)